

Income and expenditure

Macroeconomics is concerned with economy as a whole. It is thus concerned with aggregate demand and aggregate supply. By aggregate demand we mean the total amount of spending in the economy whether by consumers, by the government or by firms when they buy capital equipment or stock up on raw materials. By aggregate supply we mean the total national output of goods and services.

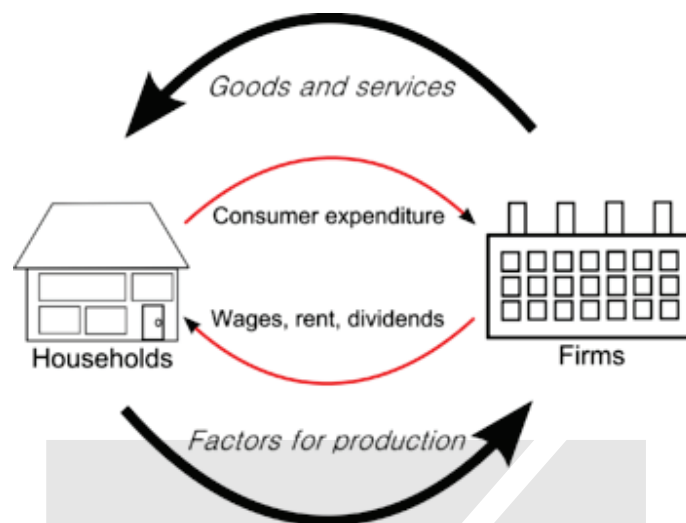
Macroeconomics studies the determination of national output and its growth over time. It also studies the problems of recession, unemployment, inflation, the balance of payments, the cyclical instability and the policies adopted by governments to deal with these problems. Macroeconomic problems are closely related to the balance between aggregate demand and supply. If aggregate demand is too high relative to aggregate supply, inflation (general increase in the price level) and balance of payment deficits (when imports expenditure exceed export revenue or payment exceeds receipts) are likely to result. If on the other hand, AD is too low relative to AS, unemployment (where supply of labor is greater than demand for labor) and recession (where output and income level fall) may well result.

Macroeconomic policy, therefore, tends to focus on the balance of AD and AS. It can be demand side policy; which seeks to influence the level of spending or AD in the economy. This in turn will affect the level of production, prices and employment or it can be supply side policy which is designed to influence the level of production or AS directly; e.g. by trying to create more incentives for workers and producers.

Circular flow of income and expenditure: It explains the working of an economy in the simplest way. It explains that the output, income and expenditures of the economy are same. The circular flow of income shows all of the money coming into an economy (injections) and all of the money that goes out of an economy (leakages or withdrawals). It allows us to see the general reasons why an economy might grow and shrink in size. In this very simple model of the whole economy, it is assumed that the households own all the factors of production. They sell these factors to the firms, earning rent on their land, wages for the use of their labor, and profit and interest for the use of their capital. The firms then use these factors to produce goods and services and the households, uses the income they earned from the sale of their factors.

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So the size of an economy can be measured using the income, output or expenditure method. Notice that the three methods should give exactly the same answer. It is fairly obvious that the amount of money spent must equal the value of the goods and services that his money is spent on. Although less obvious, it should make sense that the amount of money spent will equal the income of the spenders, assuming that none of this is saved. This brings us to another key point that there is no injections into this circular flow and no leakages from the circular flow (like saving) at this stage.

Hence, income = output = expenditure.

Including leakages and injections: In this simple model, we have, so far, assumed that the system is completely closed. It would be fair to assume, though, that households will not spend all of their income, and that firms will, on occasion, invest in new capital. Leakage will include savings and injection will include investment.

Now there are two types of expenditure: Consumption by households and investment by firms, so $AE = C + I$. also, households income is not all spent anymore, some of it is saved, so $Y = C + S$. We know that in equilibrium $Y=E$, so by substituting we have:

$$Y = E$$

$$C + S = C + I$$

$$S = I \text{ (by cancelling out the } Cs)$$

The 3-sector model: In the real world, we know that there are more 'players' in an economy than simply households and firms. The 3-sector model includes the government sector. For the purposes of the circular flow, governments do two things: they tax businesses and consumers, and they spend this money on consumers (benefits and pensions) and businesses (subsidies).

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We now have the following situation:

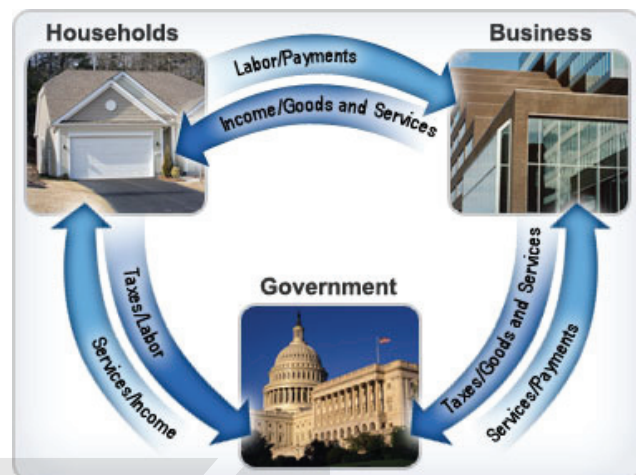
$$AE = C + I + G$$

$$Y = C + S + T$$

$Y = AE$ in equilibrium, so

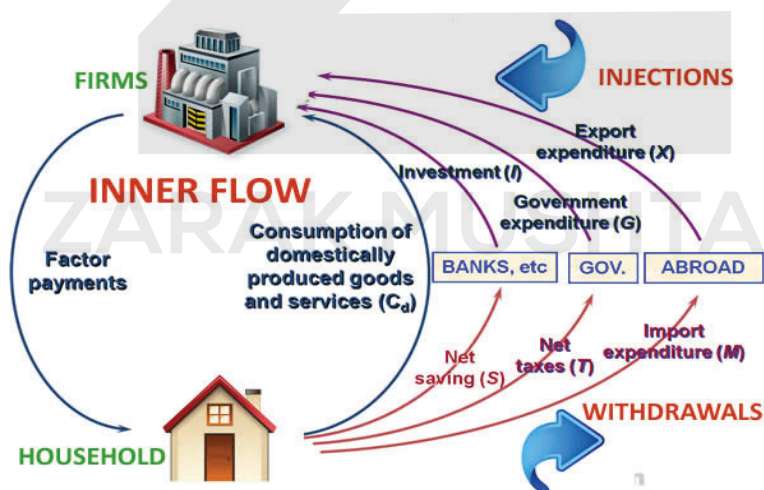
$$C + S + T = C + I + G$$

$$S + T = I + G \text{ (by cancelling the } Cs)$$



The situation is a little more complicated now. We have two leakages (saving and taxation) and two injections (investment and government spending). Now that we have a situation where actual saving does not necessarily have to equal actual investment. Now, saving and taxation together have to equal investment and government spending together. This means that investment can be greater than saving as long as taxation is higher than government spending.

The 4-sector model: X denotes export revenue and M denotes import expenditure. The foreign sector box has been added. The line of imports comes out of the consumption line. This is because it is the consumers who buy these imports which means that money leaks out of the economy. The injection into the economy is the exports.



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$$Y = AE$$

$$C + S + T + M = C + I + G + X$$

$$S + T + M = I + G + X \text{ (by cancelling out the } C\text{s)}$$

S , T and M are the leakages from an economy and I , G and X are the injections into an economy. The economy will only be in equilibrium if injections equal leakages.

National expenditure, or aggregate planned expenditure, is equal to $C + I + G + (X - M)$. The reason why M is included, but not S or T is that imports are a sub-group of consumption. C includes all consumption by UK consumers, including the consumption of imports as well as home produced goods. This has to be taken away as it is a leakage.

Gross domestic product (GDP): It is the value of final goods and services produced by domestically located factors of production; no matter where the owners are located i.e. it is the total output produced within the boundaries of a country. For example the GDP of Pakistan is the value of output produced by the domestic producers and the value of output produced by the foreign multinational companies located in Pakistan.

Gross national product (GNP): It is the value of final goods and services produced by the domestic factors of production no matter where they are located.

Final goods are goods that are ultimately consumed rather than used in the production of another good. For example, a car sold to a consumer is a final good; the components such as tires sold to the car manufacturer are not; they are intermediate goods used to make the final good. The same tires, if sold to a consumer, would be a final good. Only final goods are included when measuring national income. If intermediate goods were included too, this would lead to double counting; for example, the value of the tires would be counted once when they are sold to the car manufacturer, and again when the car is sold to the consumer. Only newly produced goods are counted. Transactions in existing goods, such as second hand cars, are not included, as these do not involve the production of new goods.

Some of the domestic output is produced by the foreign owned resources. They send their incomes to their own countries in the form of salaries, interest and profits. Similarly there is an inflow of the same type of incomes into the domestic country. The difference of the income paid abroad and the income earned abroad is called 'net property income from abroad'. If this item is added to GDP we arrive at the GNP figure.

Thus, $GNP = GDP + \text{net property income from abroad}$. For e.g. in the case of a German owned car factory operating in the US, the profits for the factory would be counted as part of German GNP rather than US GNP because the capital used in production (the factory, machinery, etc.) is German owned. The wages of the American workers would be part of US GDP.

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- $GNP = GDP + \text{Net property income from abroad}$
- $N.I = GNP - \text{depreciation}$
- $\text{Personal income (PI)} = N.I - \text{retained profits of businesses.}$
- $\text{Disposable personal income (DPI)} = P.I - \text{taxes on incomes} + \text{transfer payments.}$

Actual and potential GDP/income: Actual income represents what the economy actually produces and is denoted as 'y'. Potential income measures what economy could produce if all the resources were fully employed at their normal rated of utilization. It is denoted as 'Yf'. The gap between actual and potential GDP is called GDP gap. If potential income exceeds the actual income, the gap is called recessionary gap. But if the actual income exceeds the potential income, the gap is called the inflationary gap.

Nominal and real NI: Nominal national income sometimes called 'money national income', measures national income in terms of current prices and thus takes no account of inflation i.e. Current price level $\times$ current output.

Real income, however, measures national income in the prices that existed in some particular year - the base year. Thus, we could measure each year's national income in, say 1990 prices. This would enable us to see how much real income had changed from one year to another. In other words, it would eliminate increases in money national income that were merely due to an increase in prices.

To calculate the real GDP, we use **GDP deflator** which is an index of the average prices of all the components of GDP. Thus, it includes not just consumer goods prices, but also the prices of investment goods, goods and services consumed by the government and the prices of exports. Essentially, GDP deflator is an adjustment for the impact of changes in prices on changes in nominal GDP. GDP deflator can be considered the most comprehensive measure of inflation since a wide array for goods and services are included in its construction. But it may not reflect the full impact of inflation on consumer welfare because it does not include imported goods and services that constitute a significant portion of what people buy.

GDP deflator is the ratio of the value of aggregate final output at current market prices (nominal GDP) to its value at the base year prices (real GDP). In effect the basket of goods for the construction of this price index includes all the final output produced within the geographic boundaries of the country.

$$\text{GDP deflator} = \frac{\text{GDP at current prices}}{\text{GDP at base year prices}} \times 100$$

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FACTOR COST AND MARKET PRICE: Market price is the price of the commodity which the consumers pay. It includes taxes and subsidies. In cases where taxes are imposed, the producer receives less than the consumer pays and where subsidies are provided by the government, the producer receives more than the consumer pays.

The term factor cost refers to the value of output measured in terms of the payments made to the factors of production for the services rendered in producing the output. These payments are in the form of wages, rent, interest and profits.

Using market prices as measures of the value of output can be misleading when market prices do not accurately reflect the cost of production. Nowadays, most market prices contain some element of taxation and a few of them include the element of subsidy. In order to arrive at the factor cost value, taxes on expenditures must be deducted and subsidies must be added to the market price valuations. Thus, factor cost = market price - taxes + subsidies.

THE THREE WAYS OF MEASURING GDP: GDP can be calculated in three different ways, which should all result in the same figure. The first method of measuring GDP is to add up all the value of all the goods and services produced in the country, industry by industry. The first method is called output method.

The production of goods and services generates incomes for households in the form of wages and salaries, profits, rent and interest. The second method of measuring GDP therefore is to add up all these incomes. This is known as the income method.

The third method focuses on the expenditures necessary to purchase the nation's production. The national output must either be bought for use or added to stocks. If we assume that net additions to stock amount to expenditures by the firm on its own output, we can measure national income by the amount of money spent on purchasing the national output. This is known as the expenditure method.

Because of the way the calculations are made, the three methods of calculating GDP must yield the same result. In other words:

NATIONAL OUTPUT = NATIONAL INCOME = NATIONAL EXPENDITURE

The output method: In this method we calculate national income by adding the outputs of all the final goods and services produced in an economy during a year. But we cannot just add up all the outputs of individual firms because outputs of many firms are the inputs of many other firms. If we add up the market value of outputs of all the firms we would obtain an output in excess of the actual output. This is called double counting and in order to avoid double counting we either make use of value added method i.e. how much value has been added at each stage of production or we simply take the value of the final goods and services. For example, if a manufacturer sells a

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television to a retailer for \$200 and the retailer sell it to the consumer for \$300. We do not add the \$200 received by the manufacturer to the \$300 received by the retailer because this would lead to double counting. Instead we just count the final value i.e. \$300 or the value added at each stage i.e. \$200 by the manufacturer + \$100 by the retailer.

Exports are included as part of national income, but imported materials and services are excluded. If the value added method is used, imports will automatically be excluded since we only record the values added in this country. This will give us the GDP and to this total must be added the net property income from abroad to calculate the GNP. Finally the national income figures are adjusted for inflation because it leads to an appreciation of stocks.

Income method: In this method N.I is calculated by adding up all the incomes which have been earned for services rendered, these income are mainly divided in to four categories i.e. income from employment, income from self-employed, rent and profits.

Income from employment includes wages and salaries and they are measured gross i.e. before taxes are paid because this measure of the factor's contribution.

Income from self-employed includes incomes of those who are earning a living selling their services or output but who are not employed by any one organization e.g. consultants.

Rent is the payment for the services of land and other factors that are rented.

Profits are the incomes of the businesses. Some profits are paid out as dividends to the owners of firms and the rest are retained for use by firms. Both distributed and undistributed profits are included in the calculation of national income. A sizeable proportion of personal income is made up of transfer payments which are in form of social security payments such as unemployment benefits, old age pensions etc. these are the payments against which no output of goods and services are produced, so they are not included as part of national income. Net property income from abroad is then added to the national income account. Finally the national income is adjusted for inflation.

Expenditure method: To determine national income through expenditures, we add up all types of spending on finished or final goods and services. These expenditures are divided as: consumption, government, investment and expenditures on exports and imports.

Consumption expenditure (C) includes expenditure by households on durable consumer goods (automobiles, refrigerators, and DVD players), non-durable consumer goods (bread, milk, shirts, and pencils) and consumer services (of lawyers, doctors, and mechanics). Investment expenditures (I) refers to all the spending by firms. It includes: all final purchases of machinery, equipment and tools by business enterprises, all construction and changes in inventories. Construction includes buildings, factories, warehouses etc. Apartment buildings and houses are investment goods inventories are an unconsumed output which is nothing but investment.

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Government expenditures (G) include all the governmental spending on the finished products of businesses and all direct purchases of resources. It excludes all government transfer payments, because such outlays do not reflect any current production.

Total domestic expenditure includes spending on foreign goods and services (imports) which do not generate factor income at home. On the other hand it does not include expenditure by foreigners on domestically produced goods and services (exports) which do generate income at home. Thus exports must be added and imports subtracted from total domestic spending i.e. net exports ($X-M$) are added.

Need for the study of national income: A national income measure serves various purposes regarding economy, production, trade, consumption, policy formulation etc. The following are some such needs.

- To measure the size of the economy and level of country's economic performance
- To trace the trend or speed of the economic growth in relation to previous year(s) as well as to other countries
- To know the structure and composition of the national income in terms of various sectors and the periodical variations in them
- To make projection about the future development trend of the economy
- To help government formulate suitable development plans and policies to increase growth rates.
- To fix various development targets for different sectors of the economy on the basis of the earlier performance
- To help business firms in forecasting future demand for their products
- To make international comparison of people living standards.

PROBLEMS IN CALCULATING NATIONAL INCOME: The measurement of national income encounters many problems. The problem of double counting has already been noted. Though there are some corrective measures, it is difficult to eliminate double counting altogether. And there are many such problems and the following are some of them.

1. **Black money:** In countries where level of illegal activities, illegal businesses and the level of corruption are very high, the circulation of black money is so high, it has created a parallel economy. It means unreported economy which is equivalent to the size of officially estimated size of the economy. GDP does not take into account the parallel economy as the transaction of black money is not registered.
2. **Non-monetization:** In most of the rural economy, considerable portion of transactions occurs informally and they are called as non-monetized economy. The presence of such non-monetary economy in developing countries keeps the GDP estimates at a lower level than the actual.

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3. *Growing service sector:* In recent years, the service sector is growing faster than that of the agricultural and industrial sectors. However, value addition in legal consultancy, health services, financial and business services and the service sector as a whole is not based on accurate reporting and hence underestimated in national income measures.
4. *Household services:* The national income analysis ignores domestic work, and housekeeping and social services. Most of such valuable work rendered by our women at home does not enter our national accounting.
5. *Social services:* It ignores volunteer and unpaid social services. For example, the wonderful services of Abdul Sattar Edhi is invaluable for millions of poor, destitute, orphans and the diseased but at the same time not included in Pakistani GDP.
6. *Environmental cost:* National income estimation does not distinguish between Environmental friendly and environmental hazardous industries. The cost of polluting industries is not included in the estimate.

NATIONAL INCOME AND THE STANDARD OF LIVING

The standard of living is a measure of the material welfare of the inhabitants of a country, so income is the most important determinant of the standard of living, so movements in the national income are often used to indicate changes in the standard of living. Increase in the nominal national income may not necessarily increase the standard of living because it is the increase in the physical quantities of goods and services which directly increase the standard of living. Therefore, nominal national income has to be adjusted for inflation in calculating real national income. Account must also be taken of changes in population because it is income per head which is relevant when living standards are calculated. A 5% increase in real income which is accompanied by an 8% increase in the population probably means that the average standard of living has fallen.

National GDP data can hide regional variations in output, employment and income per head of the population, so the national income figures will not give the true picture of the standard of living of the economy as a whole.

It is also necessary to take note of the composition of total output. A large increase in total output which is due to an increased output of capital goods or defense equipment will not mean any immediate increase in economic welfare. National income figures are a measure of the material standard of living rather than the quality of life. They measure the quantity of goods and services but the quality of life is affected by many factors. If increase in the quantity of goods and services are offset by falls in their quality the standard of living will not have increased.

Higher national income figures may also give a misleading indication of changes in the economic welfare if they have occurred at the same time that working hours have increased, working conditions declined and/or negative externalities have been created.

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Externalities both negative and positive are not accounted for in the calculation of national income. Pollution, noise, congestion and mental strain may be the by-products of a rapidly increasing national income.

Output may increase but if the distribution is uneven, only a few may experience a rise in their material living standards. It is also possible that official national income figures may understate the quantity of goods and services which are produced and sold not recorded in the official figures due to the expansion of the shadow (or underground) economy and also the value of unpaid work done by millions of volunteers and people caring for their families. So the national income figures understate the true standard of living in the economy. Various definitions are used to describe the "shadow economy" but the definition usually embraces a range of illegal activities such as drug production and distribution, prostitution, theft, fraud, and concealed legal activities such as tax evasion or otherwise- legitimate business activities such as unreported self-employment income. The scale of the "shadow economy" varies widely across countries at different stages of development. According to the IMF, in developing countries it may be high as 40% of GDP; in transition countries of central and Eastern Europe it may be up to 30% of GDP and in the leading industrialized countries of the OECD, the shadow economy may be in the region of 15% of GDP.

Comparison of GDP per head and living standards between countries

The citizens of a country with a higher GDP per head are likely to enjoy higher living standards than people living in a country with a lower real GDP per head, but this is not necessarily the case.

To compare living standards between countries, it is necessary to convert the real GDP per capita into a common currency. To avoid the comparison being distorted by exchange rate changes, economists usually adjust exchange rates to take into account their purchasing power parities (PPPs). For example, suppose the exchange rate is 6 Malaysian ringgits equals \$1 and the USA has a real GDP per head of \$50000 while Malaysia has a real GDP of 12 000 ringgits. From this information, it might appear that when Malaysia's real GDP is converted into dollars (\$2000) people in the USA are, on average, 25 times better off than people in Malaysia. However, if \$1 can buy more goods and services in the USA than in Malaysia, then using the exchange rate to convert ringgits into dollars will exaggerate Malaysia's output. A basket of products may sell for \$200 in the USA and for 2400 ringgits in Malaysia. This would mean that in terms of ability to buy products (purchasing parities) 12 ringgits are worth \$1. Using this as the basis for converting Malaysia's output into US dollars would show that people in the USA are 50 times better off than people in Malaysia.

Each year, The Economist magazine publishes a 'Big Mac' index. This is a simplified version of the purchasing power parity between the \$ and a number of other currencies. It divides the price of a Big Mac in one country by the price in another country to get the purchasing power parity between the two currencies. For instance, if the price in the USA is \$4 and it is £3 in the UK, the

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purchasing power parity would be \$ 4 £ 3, so £1 = \$1.33. If the actual exchange rate is £1 = \$1.50, it would suggest that the pound is overvalued and may fall in value in the future.

Even if a country is found to have a higher real GDP per head than another country using purchasing power parities, it does not necessarily mean that its population will enjoy higher living standards. For example, Kuwait has one of the highest real GDP per head in the world, but some immigrant workers in the country receive relatively low wages. Where income is very unevenly distributed, only a small number of households may benefit from a high average income.

When assessing living standards between countries, consideration must be given to factors that are not measured in real GDP per head, just as when making comparisons over time. For example, working hours, working conditions, fear of crime and freedom of thought may be taken into account.

Human development index (HDI)

It combines a number of development indicators into one index with the maximum possible value of 1. Changes in the index over time can therefore show whether a country has improved or reduced the economic well-being of its people in terms of their:

- Standard of living as measured by gross domestic (GDP) per capita
- Access to education and knowledge, measured by the adult literacy rate and school and college enrollment rates.
- Health, diet and lifestyle, measured by life expectancy at birth; longevity.

Measurable Economic Welfare

Another composite measure is Measurable Economic Welfare (MEW). This was developed in 1972 by two American economists, William Nordhaus and James Tobin. The measure seeks to give a fuller picture of living standards by adjusting GDP figures to take into account other factors that have an impact on living standards. Factors that improve living standards, including increased leisure time, are added to the GDP figure. Factors that reduce living standards, such as environmental damage, are deducted. Of course, in practice, it is difficult and expensive to measure the value of non-marketed goods and services.

Multidimensional Poverty Index

A more recent composite measure is the Multidimensional Poverty Index (MPI). This was developed in 2010 by the Oxford Poverty and Human Development Initiative and the United Nations Development. It measures indicators of living standards (cooking fuel, sanitation, drinking water, electricity, housing and assets), education (years of schooling and school attendance) and health (child mortality and nourishment). The six indicators of living standards are given a total weighting of 33%, the two indicators of education a total weighting of 33% and similarly the two indicators of health have a weighting of 33%. A household is considered to be multidimensional poor if they are

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deprived in at least 33% of the weighted indicators. This means a family would be regarded as poor if it has lost a child and has another child who is not attending school.

The aim of the MPI is to help countries understand why people are poor and why some stay poor even when incomes rise. The MPI can help governments and international organizations target the poorest groups in a country, assess and co-ordinate national development plans.

The Kuznets Curve: The Kuznets curve is a U-shaped curve that was first drawn by the American economist Simon Kuznets. The Kuznets curve suggests that as an economy develops, income becomes more unevenly distributed, and then after a certain income level is reached, income becomes more evenly distributed as shown in Figure 50.6.

The reason why it is thought there will be an initial rise in income inequality is that as an economy develops there will be a movement of labour from low-paid and low-skilled agricultural jobs to higher-paid and more skilled manufacturing jobs.

However, this suggested trend is not being followed in all developing economies. For example, the curve does not explain the rise in income inequality in the high-income economy of the USA.

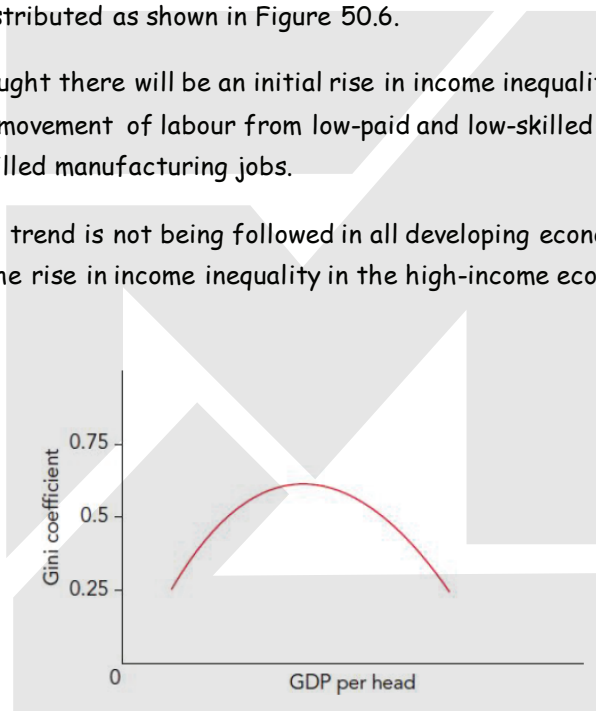


Figure 50.6: The Kuznets curve

ZARAK MUSHTAQ

National income

The theory of national income determination can be best understood by making use of simple Keynesian model of an economy. This model is named after the great English economist John Maynard Keynes who was one of the main founders of modern macroeconomics. He developed the model in his famous book, *The General Theory of Employment, interest and money*. The Keynesian model describes the economy's short run behavior which is mainly determined by fluctuations in the demand for the nation's output, i.e. its GDP.

Keynesians argue that aggregate demand determines the level of economic activity in the economy. In other words, the nation's production and employment depend on the amount of spending. Too

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little spending will lead to unemployment. More spending will stimulate firms to produce more and employ more. Too much spending however, will cause inflation.

In the simple Keynesian theory, the consumption of domestically produced goods (C) and the three withdrawals (W) = saving (S), net taxes (T) and spending on imports (M) - all depend on the level of national income (Y). In fact, in the model, national income must always equal consumption of domestic goods plus withdrawals.

$$Y = C + W$$

Total spending in the economy on the goods and services of domestic firms is called aggregate demand (AD). It is also referred to as national expenditure on domestic product (AE). It consists of C plus the three injections (J): investment in the domestic economy (I), government expenditure in the domestic economy (G) and expenditure on exports (X).

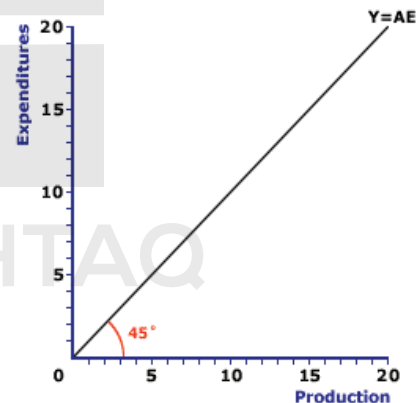
$$AD = AE = C + J$$

In equilibrium, withdrawals equal injections. Since national income (Y) is simply withdrawals plus C and aggregate expenditures (E) is simply injections plus C , it follows that in equilibrium national income must equal aggregate expenditures.

$W = J$	when $AE > Y$	$J > W$	$Y \uparrow$	$S \uparrow$	$M \uparrow$	$T \uparrow$	$(W \uparrow)$ until $W = J$
$C + W = C + J$	when $AE < Y$	$J < W$	$Y \downarrow$	$S \downarrow$	$M \downarrow$	$T \downarrow$	$(W \downarrow)$ until $W = J$

$$Y = AE (=AD)$$

Whenever aggregate expenditures ($C + J$) exceed national income ($C + W$), injections will exceed withdrawals. Firms will respond to the extra demand by producing more and hence employing more factors of production. National income will thus rise. But as national income rises, so too will saving, imports and the amount paid in taxes; in other words withdrawals will rise. Withdrawals will go on rising until they equal injections; until a new equilibrium has been reached.



In this model, it is assumed that the levels of consumption and withdrawals are determined by the level of national income i.e. they are induced. Since, national income is part of the model; we say that consumption and withdrawals are endogenous. Injections, however, are assumed to be exogenous: they are determined independently of the level of income i.e. they are autonomous expenditures.

The 45 degree line: We plot real national income (RNI) on the horizontal axis and the various component parts of aggregate expenditure (AE) on the vertical axis. The 45 degree line out from

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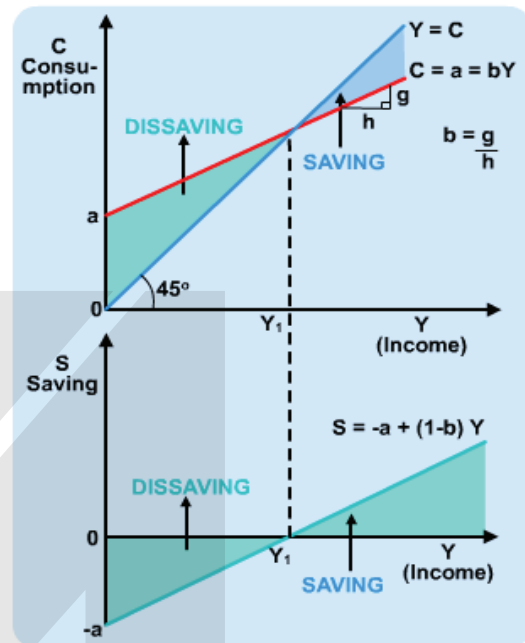
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the origin plots $Y (C + W)$. When the same scale is taken on both the axes, all the points to the 45 degree line show equal AE and Y levels in the economy.

The consumption function (c):

The consumption function refers to a positive relation between income and consumption. It is fundamental relationship between two aggregate i.e. C and Y . Symbolically, the relationship is represented as: $C = f(Y)$, where C is a dependent variable and Y is an independent variable i.e. C is determined by Y . Thus, consumption function is a schedule of the various amounts of consumption expenditure corresponding to different levels of income. This relationship is based on the *Ceteris paribus* assumption i.e. all other possible influences on consumption are held constant.

A typical Keynesian linear consumption function is represented as:



$C = C + cY$, where C is the autonomous consumption which does not depend on national income. When $Y = 0$, a typical individual will still consume some minimum amount through borrowing or drawing down savings. Many individuals have no income but they continue to consume e.g. dependent children or non-working spouses. cY is the induced part of consumption i.e. an increase in income will lead to increase in consumption by some proportion c .

The marginal propensity to consume (mpc): The MPC may be defined as the ratio of the change in consumption to the change in income i.e. it is the proportion of any increase in the national income that goes on consumption.

MPC = change in C / change in Y

MPC shows the slope of consumption function and the value of MPC ranges between 0 and 1 which means when increase, the whole of it is not spent on consumption e.g. if $MPC = 0.8$, it means that 80% of the increased income will be spent as consumption expenditure.

Average propensity to consume (APC): It measures the part of income that will become the consumption expenditure. It is calculated as the total consumption expenditures divided by real disposable income.

APC = C / Y_d

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The APC declines as income level rises because the proportion of income spent on consumption decreases. The higher the level of Y , the smaller the proportion that will be consumed i.e. APC fall.

Using the concepts of APC and MPC we can summarize the properties of short-run consumption function as:

1. There is a break-even level of income at which $APC = 1$. Below this level $APC > 1$ i.e. $C > Y$, so consumers run down savings or borrow. Above the break-even level $Y > C$, so there is positive saving.
2. $MPC > 0$ but < 1 for all levels of income. This means that for each additional \$1 of income, less than \$1 is spent on consumption and the rest is saved. For a straight line consumption function the MPC is constant at all levels of income.

The saving function: Saving is the residual part of income after consumption expenditures have been done i.e.

$$Y = C + S$$

$$S = Y - C$$

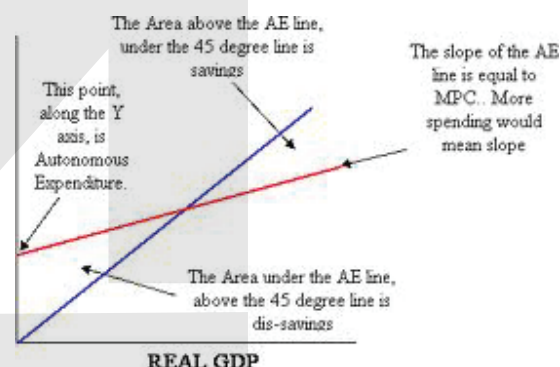
$$S = Y - (C + cY)$$

$$S = Y - C - cY$$

$$S = -C + Y - cY$$

$$S = -C + (1 - c)Y$$

$$S = -C + sY$$



Where s = marginal propensity to save and $-C$ shows the dissaving which arise due to the positive autonomous consumption expenditures.

Marginal propensity to save is the proportion of additional income that goes to saving. MPS shows the gradient of saving function which is equals to $1 - mpc$ as $mps + mpc = 1$.

MPS related the change in total desired saving to change in disposable income.

Average propensity to save is the part of the total income that will become saving. It is calculated as the total saving divided by total income.

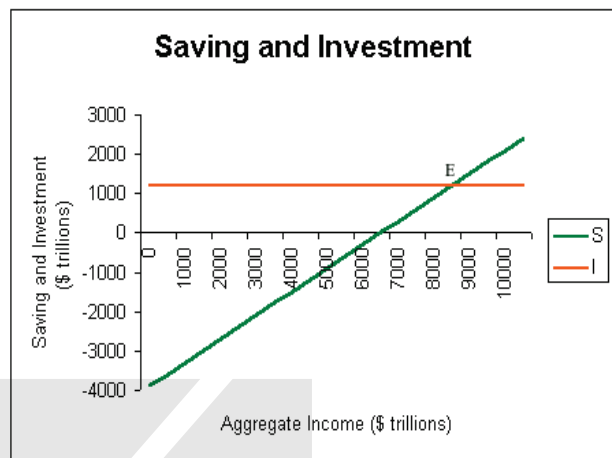
$$APS = S/Y_d$$

APS increases with the increase in income because the proportion of income that goes to saving tends to rise. Since income is either spent or saved, it follows that the fractions of income consumed and saved must equal to total income. If $mpc = 0.8$, then mps will be 0.2 .

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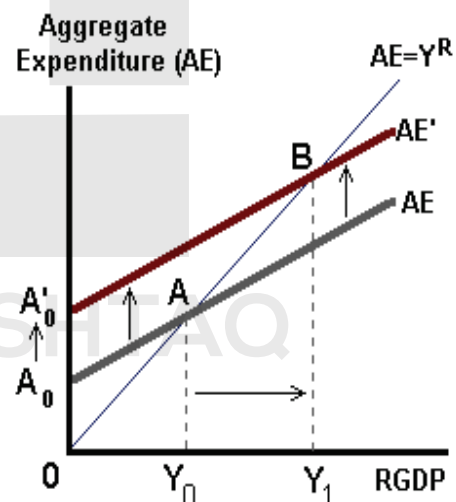
Income and expenditure

Desired investment function: Investment expenditure is the most volatile component of GDP, and changes in investment expenditures are strongly associated with economic fluctuations. Investments primarily depend on real rate of interest. Other things being equal, the higher the real interest rate, the higher is the cost of borrowing money for investment purposes and the less is the amount desired investment expenditure. Some of the other factors which influence investments are; business expectations, consumer demand, the level of income, existing stock of capital etc.



In this simple Keynesian model, we treat investment as autonomous, i.e. it is uninfluenced by the level of national income. This will help us to study how national income is determined when there is an unchanged amount of desired investment expenditure and how changes in investment cause changes in national income. Investments in economic and social overheads whether made by the government or the private sector is autonomous. Such investment includes expenditures on buildings dams, roads, canals, schools, hospitals, etc. Since investment on these projects is generally associated with public policy.

The multiplier: When autonomous expenditure change, the AE function shifts causing a resultant change in real national income. This change in real national income is a multiple of the initial increase in autonomous expenditure i.e. the change in RNI is multiplier times the change in autonomous aggregate expenditure. The multiplier is the ratio of the change in RNI to the change in autonomous AE i.e. it shows the magnitude by which RNI will rise.



$$\text{Multiplier (m)} = \frac{1}{1-\text{MPC}} = \frac{1}{\text{MPS}}$$

$$\text{Total increase in National Income} = \text{Initial Injection} \times \text{Multiplier}$$

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Income and expenditure

$$Y = C + I$$

$$Y = C + cY + I$$

$$Y - cY = C + I$$

$$Y(1 - c) = C + I$$

$$Y = 1 / (1 - c) \times AE \quad \{C + I = AE\}$$

$$Y = \infty AE \quad \{\text{multiplier} = \infty = 1/1-c \text{ or } 1/mps\}$$

If $mpc = 0.8$, then the value of multiplier will be $\frac{1}{1 - 0.8} = 5$

The value of multiplier depends on mpc . The greater the value of mpc , the greater will be the value of multiplier and vice versa.

Multiplier times increase in national income due to increase in investment:

An increase in investment will lead to a multiple increase in national income. This is called the multiplier effect and is an important part of Keynesian theory of income and employment.

The value of multiplier depends on the slope of AE function i.e. on marginal propensity to consume (mpc) or marginal propensity to spend. In a 2 sector model, where there are consumption and investment expenditures, the value of multiplier depends on mpc . The higher the value of mpc , the higher will be the multiplier and vice versa.

$$Y = (1/1 - c) \times AE \text{ where, multiplier is } = (1/1 - c)$$

The multiplier process can be understood by assuming a two sector economy in which the $mpc = 0.8$ and there is an initial increase in autonomous expenditure, say investment, of \$100m. This initial injection of \$100m will generate an immediate N.I of \$100. When this income is received by household, 80% will be spent on goods and services of domestic firms and 20% will be saved / withdrawn. This increase in consumption thus generates additional incomes of \$80m over and above the initial \$100m. Again 80% of \$80m i.e. \$64m will be spent on domestic output and \$16m will be saved and this goes on until the additional income reaches zero. Given that $mpc = 0.8$, the total increase in Y can be calculated as

$$Y = 1/ (1-0.8) \times \$100 = Y = 5 \times \$100 = \$500$$

So an initial increase in investment expenditure of \$100 will lead to a 5 times increase in Y i.e. by \$500.

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Income and expenditure

What is the multiplier process?

- An initial change in aggregate demand can have a much greater final impact on equilibrium national income
- This is known as the **multiplier effect**
- It comes about because **injections of new demand** for goods and services into the **circular flow of income** stimulate further rounds of spending – in other *words* "one person's spending is another's income"
- This can lead to a bigger eventual effect on output and employment

Examples of the multiplier effect at work

- Consider a **£300 million increase in capital investment** – for example created when an overseas company decides to build a new production plant in the UK
- This may set off a chain reaction of increases in expenditures. Firms who produce the capital goods and construction businesses who win contracts to build the new factory will see an increase in their incomes and profits
- If they and their employees in turn, collectively spend about 3/5 of that additional income, then £180m will be added to the incomes of others.

At this point, total income has grown by $(£300m + (0.6 \times £300m))$.

The sum will continue to increase as the producers of the additional goods and services realize an increase in their incomes, of which they in turn spend 60% on even more goods and services.

The increase in total income will then be $(£300m + (0.6 \times £300m) + (0.6 \times £180m))$.

Each time, the extra spending and income is a fraction of the previous addition to the circular flow.

The Multiplier and Keynesian Economics

- The concept of the multiplier process became important in the 1930s when **John Maynard Keynes** suggested it as a tool to help governments to maintain high levels of employment
- This "demand-management approach", designed to help overcome a shortage of capital investment, measured the amount of government spending needed to reach a level of national income that would prevent unemployment.

Factors that affect the value of the multiplier effect

- The higher is the **propensity to consume** domestically produced goods and services, the greater is the multiplier effect. The government can influence the size of the multiplier through changes in direct taxes. For example, a cut in the rate of income tax will increase the amount of extra income that can be spent on further goods and services

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Income and expenditure

- Another factor affecting the size of the multiplier effect is the **propensity to purchase imports**. If, out of extra income, people spend their money on imports, this demand is not passed on in the form of fresh spending on domestically produced output. It leaks away from the circular flow of income and spending, reducing the size of the multiplier.

The multiplier process also requires that there is **sufficient spare capacity** for extra output to be produced - link to the concept of deflationary gap!

If **short-run aggregate supply is inelastic**, the full multiplier effect is unlikely to occur, because increases in AD will lead to higher prices rather than a full increase in real national output. In contrast, when SRAS is perfectly elastic a rise in aggregate demand causes a large increase in national output.

In short - the multiplier effect will be larger when

1. The propensity to spend extra income on domestic goods and services is high
2. The marginal rate of tax on extra income is low
3. Consumer confidence is high (this affects willingness to spend gains in income)
4. Businesses in the economy have the capacity to expand production to meet increases in demand

Time lags and the multiplier effect

- It is important to remember that the multiplier effect will take time to come into full effect
- A good example is the fiscal stimulus introduced into the US economy by the Obama government. They have set aside many billions of dollars of extra spending on infrastructure spending but these sorts of capital projects can take years to be completed. Delays in sourcing raw materials, components and finding sufficient skilled labour can limit the initial impact of the spending projects.

Calculating the value of the multiplier: The formal calculation for the value of the multiplier is

$$\text{Multiplier} = 1 / (\text{sum of the propensity to save} + \text{tax} + \text{import})$$

Determination of national income with government: Three sector closed economy: It is called a closed economy with government as the third sector

$$Y = C + I + G$$

Government spending and taxation policies affect equilibrium national income in two ways. First, government expenditure is the part of autonomous expenditures in our model i.e. expenditures which are not related with national income. Second, in deriving disposable income, taxes must be subtracted from national income and government transfer payments must be added.

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Income and expenditure

DY (disposable national income) = $Y - T + \text{transfers}$

$DY = Y - (T - \text{transfers})$

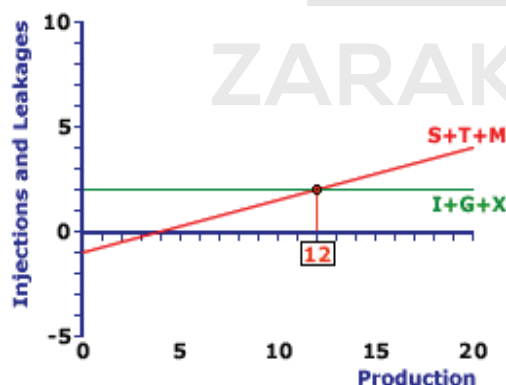
{ $T - \text{Transfers} = \text{net taxes}$ }

Net taxes are the total tax revenues received by the government minus total transfer payments i.e. net taxes is that part of national income which is left with the government. Since transfer payments are smaller than total taxes, net taxes are positive and personal disposable income is less than national income. With the taxes on incomes, the consumption then becomes the function of disposable income. Taxes reduce the gradient of the consumption function i.e. consumption will be less than before.

We assume government expenditure as autonomous i.e. G , the government is assumed to decide on how it wishes to spend and hold to these plans whatever is the level of a national income. Government can run a budget deficit (when government expenditures exceed the tax revenue) or surplus (when tax revenue exceeds the government expenditures). Moreover, the government can always raise or lower taxes if it wants to.

Four sector open economy: The 4 sector macro-economic model includes net exports (i.e. exports minus imports) as a fourth component of AE i.e. $AE = C + I + G + NX$

Exports are taken to be autonomous. This means as domestic output increase, volume of exports remain constant. The reason behind is that export is domestic output consumed by foreign citizens. If foreign income increases, the export is likely to increase. So export is dependent on foreign income and independent of domestic income. An import is foreign output consumed by domestic citizens. As domestic income rises a part of the extra income will be spent on foreign output consumed by domestic citizens. As domestic incomes rises a part of the extra income will be spent on foreign output, or there is a positive relationship between income and imports.



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